SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)
(Established under section 3 of the UGC Act, 1956)
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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

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1. Experiment Title:

To perform Data different data types (string, number, boolean, etc.) and display them.

- **a)** Learn primitive data types: string, number, boolean, undefined, null.
- **b)** Declare variables using var, let, and const.
- **c)** Understand type coercion and type conversion.

2. Software/Tools Required:

Google Chrome (Web Browser), Visual Studio Code (Code Editor), Browser Developer Tools / Console, HTML, CSS, Javascript

3. Experiment Program Code:

Experiment 2.html

```
<!DOCTYPE html>
<html>
<head>
    <meta charset="UTF-8">
    <title>Billing Calculator</title>
    <link rel="stylesheet" href="Experiment 2.css">
</head>
<body>

    <h1>Billing Calculator</h1>

    <div class="container">

        <label>Customer Name</label>
        <input type="text" id="name">

        <label>Product Name</label>
        <input type="text" id="product">

        <label>Price Per Item (Rs.)</label>
        <input type="number" id="price">

        <label>Discount (%)</label>
        <input type="number" id="discount" value="0">

        <label>Quantity</label>
        <input type="number" id="quantity">

        <button onclick="calculateBill()">Generate Bill</button>

        <div id="output">

            <h2>Billing Details</h2>
```

```
            <table border="1">
                <tr>
                    <th>Shop Name</th>
                    <td id="shop"></td>
                </tr>
                <tr>
                    <th>Customer Name</th>
                    <td id="customer"></td>
                </tr>
                <tr>
                    <th>Product Name</th>
                    <td id="productName"></td>
                </tr>
                <tr>
                    <th>Price Per Item</th>
                    <td id="priceItem"></td>
                </tr>
                <tr>
                    <th>Quantity</th>
                    <td id="qty"></td>
                </tr>
                <tr>
                    <th>Subtotal</th>
                    <td id="subtotal"></td>
                </tr>
                <tr>
                    <th>Discount</th>
                    <td id="discountValue"></td>
                </tr>
                <tr>
                    <th>Discount Amount</th>
                    <td id="discountAmount"></td>
                </tr>
                <tr>
                    <th>Amount After Discount</th>
                    <td id="afterDiscount"></td>
                </tr>
                <tr>
                    <th>GST Rate</th>
                    <td>18%</td>
                </tr>
                <tr>
                    <th>GST Amount</th>
                    <td id="gst"></td>
                </tr>
                <tr>
                    <th>Total Amount</th>
                    <td id="total"></td>
                </tr>
            </table>

        </div>

    </div>

    <footer style="text-align: center; margin-top: 30px; font-size: 14px; color: gray; padding-top:
15px;">
        Made by Dhanish Ladwani PRN - 24070521020
    </footer>

    <script src="Experiment 2.js"></script>

</body>
</html>
```

Experiment 2.css

```
body {
    font-family: Arial, sans-serif;
    background-color: lightgray;
```

```
    margin: 0;
    padding: 20px;
}

h1 {
    text-align: center;
    color: black;
}

.container {
    width: 550px;
    margin: auto;
    background-color: white;
    padding: 25px;
    border-radius: 8px;
    box-shadow: 0px 0px 10px gray;
}

label {
    display: block;
    margin-top: 12px;
    font-weight: bold;
    color: black;
}

input {
    width: 100%;
    padding: 8px;
    margin-top: 5px;
    border: 1px solid gray;
    border-radius: 4px;
    box-sizing: border-box;
}

button {
    width: 100%;
    padding: 10px;
    margin-top: 20px;
    background-color: blue;
    color: white;
    border: none;
    border-radius: 5px;
    font-size: 16px;
    cursor: pointer;
}

button:hover {
    background-color: navy;
}

#output {
    margin-top: 30px;
}

h2 {
    text-align: center;
    color: blue;
}

table {
    width: 100%;
    border-collapse: collapse;
}

th {
    background-color: blue;
    color: white;
    text-align: left;
    padding: 10px;
```

```
    width: 40%;
}

td {
    padding: 10px;
    color: black;
}

tr:nth-child(even) {
    background-color: whitesmoke;
}
```

Experiment 2.js

```
function calculateBill() {

    var customerName = document.getElementById("name").value;

    let product = document.getElementById("product").value;
    let price = Number(document.getElementById("price").value);
    let discount = Number(document.getElementById("discount").value);
    let quantity = Number(document.getElementById("quantity").value);

    const GST = 0.18;
    const SHOP_NAME = "ABC Super Store";

    let subtotal = price * quantity;
    let discountAmount = subtotal * (discount / 100);
    let amountAfterDiscount = subtotal - discountAmount;
    let gstAmount = amountAfterDiscount * GST;
    let total = amountAfterDiscount + gstAmount;

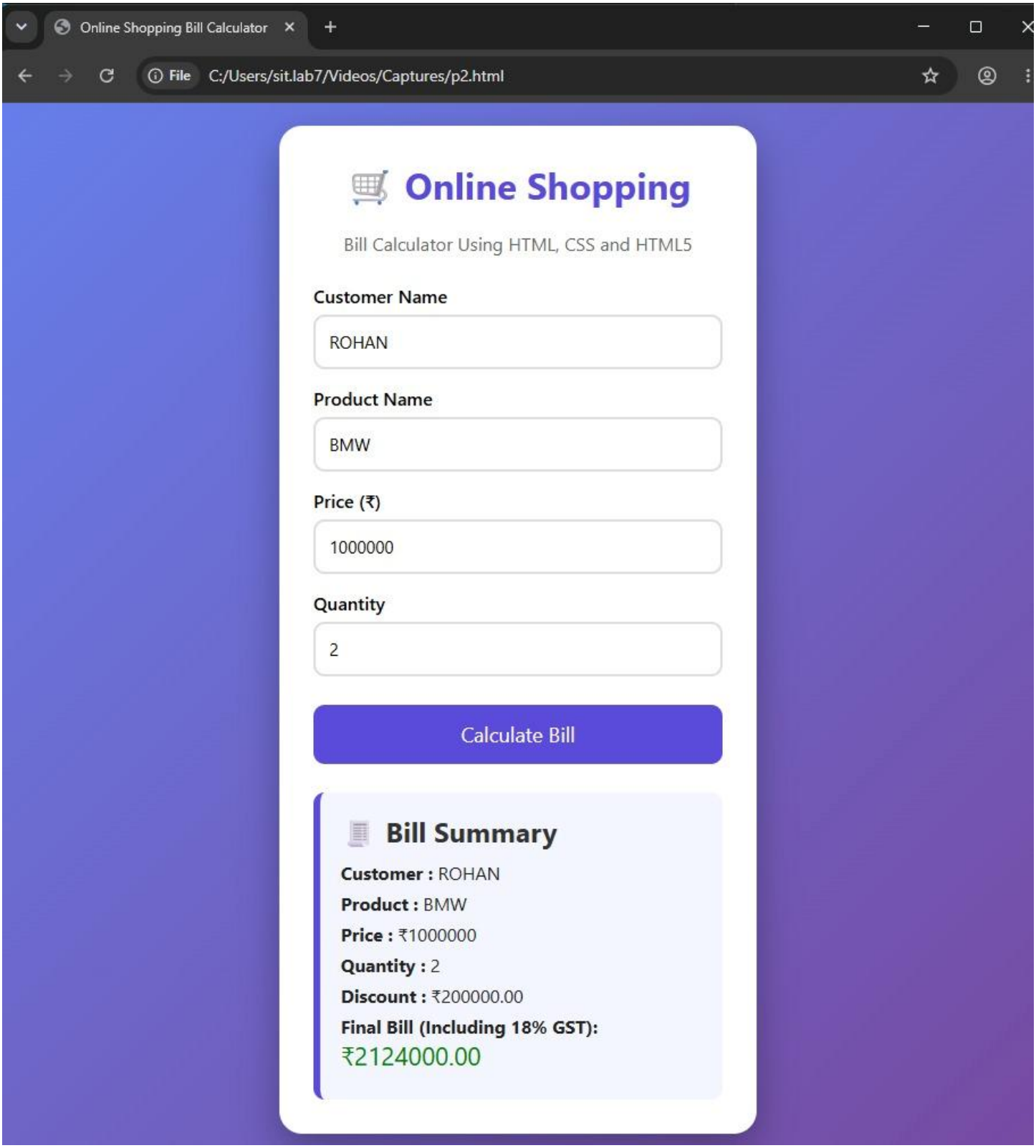
    const bill = {
        customerName,
        product,
        price,
        quantity,
        discount,
        subtotal,
        discountAmount,
        amountAfterDiscount,
        gstAmount,
        total
    };

    const {
        customerName: name,
        product: item,
        price: itemPrice,
        quantity: qty,
        discount: disc,
        subtotal: sub,
        discountAmount: discAmount,
        amountAfterDiscount: finalSub,
        gstAmount: gst,
        total: finalAmount
    } = bill;

    document.getElementById("shop").innerHTML = SHOP_NAME;
    document.getElementById("customer").innerHTML = name;
    document.getElementById("productName").innerHTML = item;
    document.getElementById("priceItem").innerHTML = "Rs. " + itemPrice.toFixed(2);
    document.getElementById("qty").innerHTML = qty;
    document.getElementById("subtotal").innerHTML = "Rs. " + sub.toFixed(2);
    document.getElementById("discountValue").innerHTML = disc + "%";
    document.getElementById("discountAmount").innerHTML = "Rs. " + discAmount.toFixed(2);
    document.getElementById("afterDiscount").innerHTML = "Rs. " + finalSub.toFixed(2);
    document.getElementById("gst").innerHTML = "Rs. " + gst.toFixed(2);
    document.getElementById("total").innerHTML = "<b>Rs. " + finalAmount.toFixed(2) + "</b>";
```

```
    console.log(`Customer: ${name}`);
    console.log(`Product: ${item}`);
    console.log(`Price Per Item: Rs. ${itemPrice.toFixed(2)}`);
    console.log(`Quantity: ${qty}`);
    console.log(`Subtotal: Rs. ${sub.toFixed(2)}`);
    console.log(`Discount: ${disc}%`);
    console.log(`Discount Amount: Rs. ${discAmount.toFixed(2)}`);
    console.log(`Amount After Discount: Rs. ${finalSub.toFixed(2)}`);
    console.log(`GST: Rs. ${gst.toFixed(2)}`);
    console.log(`Total Bill: Rs. ${finalAmount.toFixed(2)}`);
}
```

4. Output:



5. Result/Conclusion:

JavaScript's data types, variables (var, let, const) and dynamic typing were demonstrated through a billing calculator that accepts customer, product, price, discount and quantity details, and computes the subtotal, discount amount, GST (18%) and final total using object destructuring and template literals, displaying the result in a formatted table. This reinforced the concepts of primitive data types, type coercion and arithmetic operations in JavaScript.